Introduction to DAX

DAX or Data Analysis Expression is a concept that you should learn if you aspire to get the most out of Power BI capabilities or want to master it. These are a set of functions & operators which helps a user create new information out of existing data or analyze critical KPI's.

How do we structure DAX?

Similar to how we write Excel functions, DAX also has predefined functions & operators which are clubbed together or used individually to create your final DAX formula. Below is an example of a DAX function which sums two columns separately and then divide them to get the final result

```
=SUM(Column1) / SUM(Column2)
```

Here we have used the **"SUM"** function with the division operator **"/"** to get the final result

What are the different Operators used in DAX?

	Operator	Usage
Arithmetic	+	Addition
	-	Subtraction
	*	Multiplication
	/	Division
	^	Exponet
Comparison	=	Equal To
	==	Strict Equal To
	>	Greather Than
	<	Less Than
	>=	Greather Than or Equal To
	<=	Less Than or Equal To
	<>	Not Equal To
Text	&	Concatenation
Logical	&&	And
	||	Or
	IN	Logical OR

Where can we use DAX?

There are 2 ways we can use a DAX

Calculated Columns :

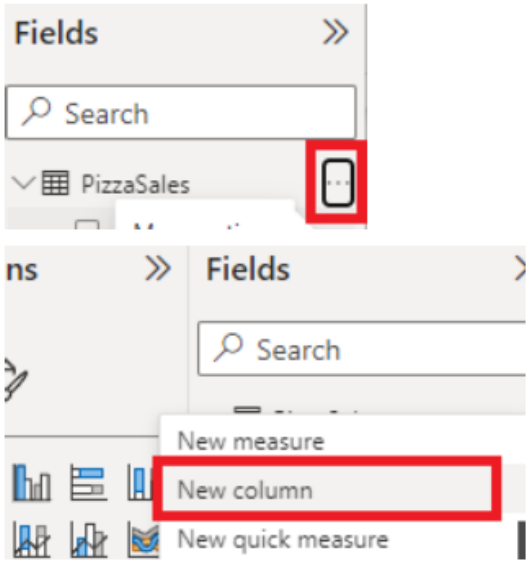
When you want your calculations to be done on a row level basis then we enter DAX into a Calculated Column. Let me explain this with an example.

Requirement

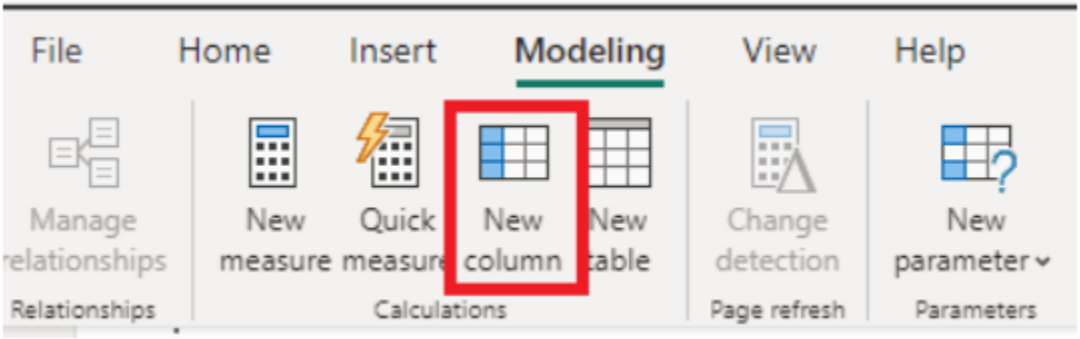
Create a new column which will show the Revenue for each row in the data using the Price_USD & Sales column

So in this case, since we need to calculate the Revenue for each row separately and store it inside a table physically, we will create a Calculated Column. Usually while creating a calculated column, I prefer creating it from the Data view because it gives us a visual preview of the data. However there are other ways as well which are listed below:

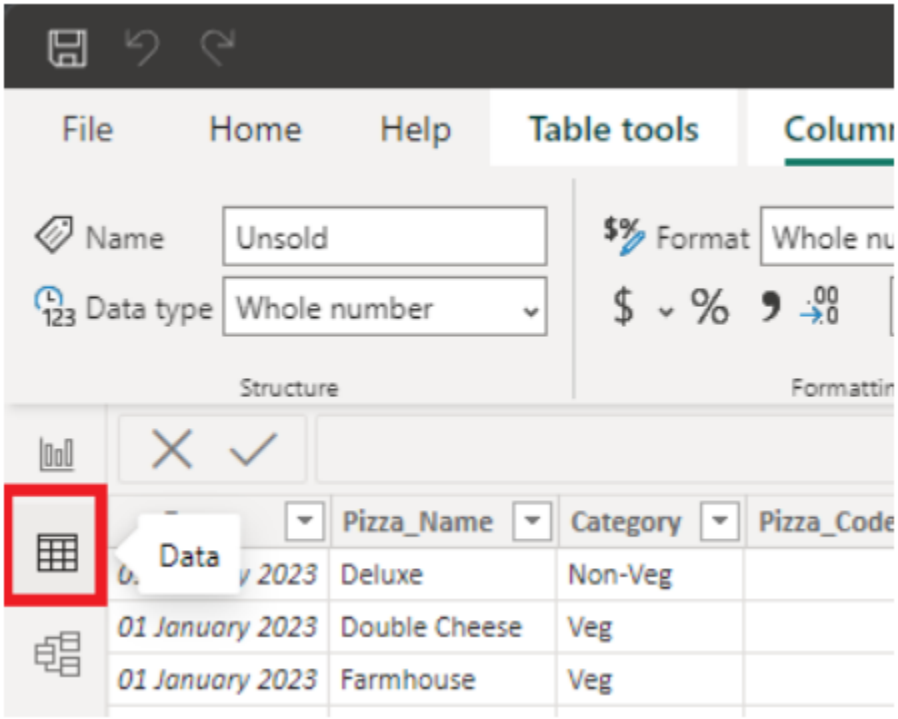
- After importing the dataset into Power BI
    - **Option 1** : Click on the 3 dots icon in the fields pane besides the Table name & click on Create



- **Option 2 :** Or click on the New Column option in the Modelling tab

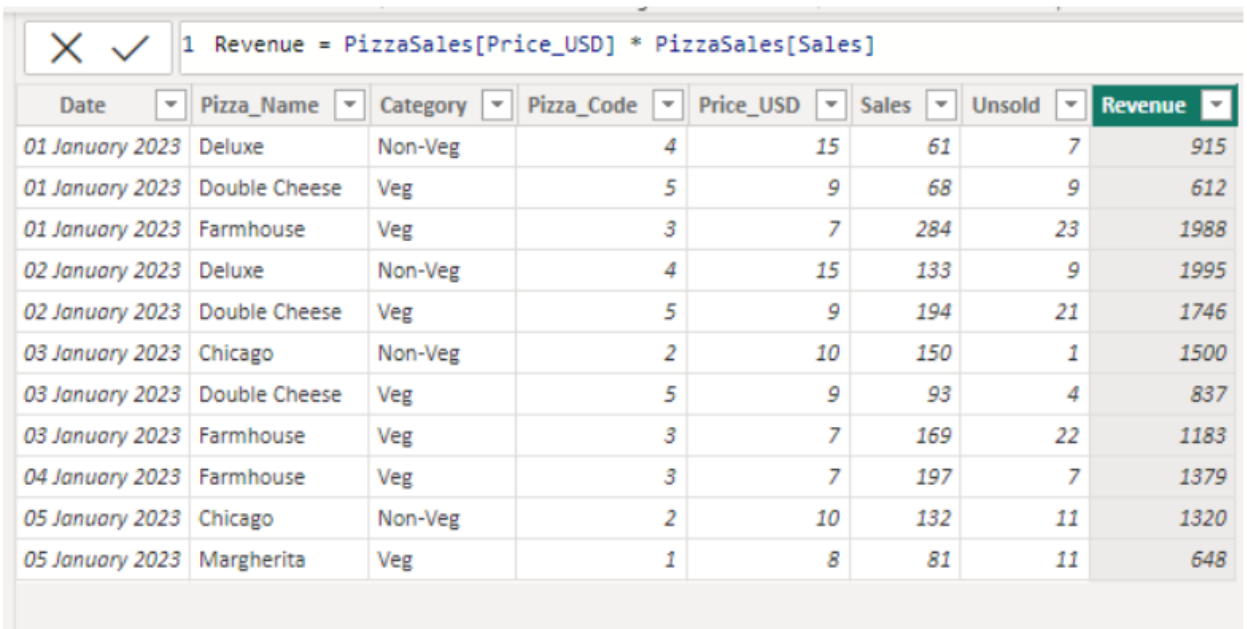


- **Preferred Option** : Or the preferred way is to go to Data view by clicking on Data View button



- Now right click anywhere on the table where you want to create the column and click **"New Column"**
- And in the formula bar enter below formula

```
Revenue = PizzaSales[Price_USD] * PizzaSales[Sales]
```



```
1 Revenue = PizzaSales[Price_USD] * PizzaSales[Sales]
```

Date	Pizza_Name	Category	Pizza_Code	Price_USD	Sales	Unsold	Revenue
01 January 2023	Deluxe	Non-Veg	4	15	61	7	915
01 January 2023	Double Cheese	Veg	5	9	68	9	612
01 January 2023	Farmhouse	Veg	3	7	284	23	1988
02 January 2023	Deluxe	Non-Veg	4	15	133	9	1995
02 January 2023	Double Cheese	Veg	5	9	194	21	1746
03 January 2023	Chicago	Non-Veg	2	10	150	1	1500
03 January 2023	Double Cheese	Veg	5	9	93	4	837
03 January 2023	Farmhouse	Veg	3	7	169	22	1183
04 January 2023	Farmhouse	Veg	3	7	197	7	1379
05 January 2023	Chicago	Non-Veg	2	10	132	11	1320
05 January 2023	Margherita	Veg	1	8	81	11	648

Calculated Measures :

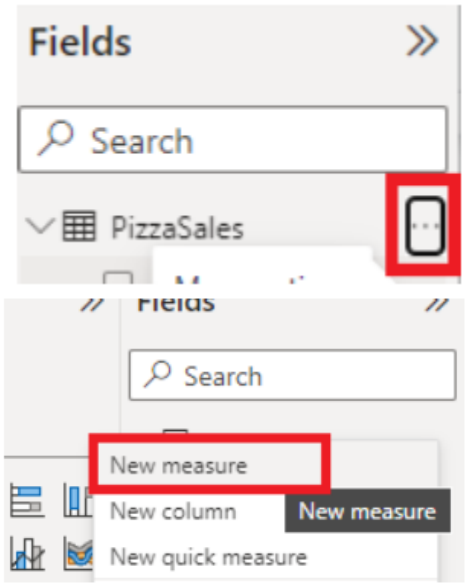
When you want to do the calculation on an aggregated level and don't particularly want to store information on the row level basis. In comparison to Columns, measures don't create physical values in your table, hence do not increase file size.

Requirement

The pizza shop ran a 15% discount on each pizza, so calculate the final revenue for all pizza sold using Price_USD, Sales & Discount%

Now in case we can create a Calculated Measure because we don't want to store the values physically for each row

- For measures I prefer to use Report view only, so click on the 3 dots icon besides the table name in Fields section and click on **New Measure**



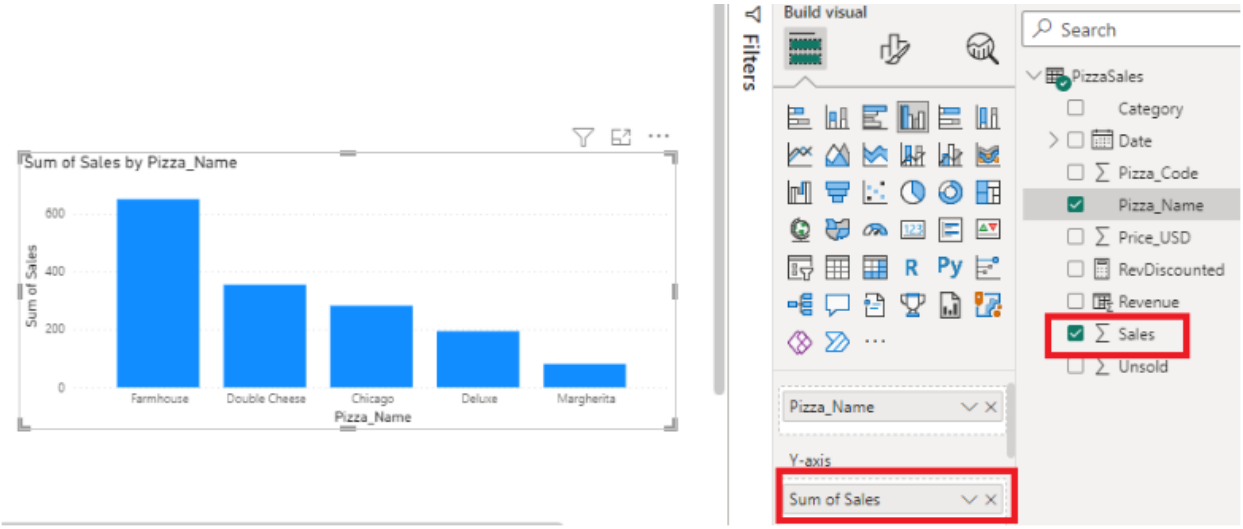
- And now in the Formula bar enter the formula below. Give any name as per your choosing, for now I have used "RevDiscounted"

```
RevDiscounted = SUMX(PizzaSales, ((PizzaSales[Price_USD] *
PizzaSales[Sales]) * 0.15))
```

Now if you have observed, since Calculated Measures are created on an aggregated level, you have to encapsulate your column name inside an aggregation function like we used **SUMX** in this case.

Implicit Measures vs. Explicit Measures

Now finally just a concept to keep in mind. By default when you drag a column into the report layout, Power BI automatically creates an aggregation measure which is known as **Implicit Measure**. Like in below case when I dragged Sales column to the Y axis to create a chart, it automatically applied a SUM aggregation on the column to give me a grouped value for every Pizza Name



On the other hand, **Explicit Measures** are the measures which we create by entering a custom formula in the formula bar like we did earlier while calculating Revenue & RevDiscounted.

Both types of measures serve their own purpose but Explicit measures are much more flexible and help in building complex logic for custom analysis on metrics & KPI's.